

A Content-Based Approach to Internship Recommendation Using TF-IDF and Cosine Similarity

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Abstract

Matching candidates to relevant internships typically requires manually comparing resumes against job requirements, which does not scale well as the number of postings grows. This report describes a small content-based recommendation system that parses resume text, extracts candidate skills using natural language processing techniques, and ranks internship postings by similarity using TF-IDF vectorization and cosine similarity. The system was implemented in Python using Scikit-Learn and exposed through a Flask interface. We describe the approach, present qualitative results from testing on a small internship dataset, and discuss the limitations of the current implementation along with directions for future work.

1. Introduction

Internship recommendation is a text-matching problem: given a candidate's skills and experience as described in their resume, identify postings whose requirements most closely align. A straightforward approach is keyword-based filtering, but this struggles with variation in how skills are phrased across resumes and postings. This project explores a content-based filtering approach using classical NLP techniques — TF-IDF vectorization and cosine similarity — to rank postings by relevance to a parsed resume, without requiring labeled training data or user interaction history.

2. Approach

Resume parsing. Resume text is parsed to extract candidate skills and relevant experience using NLP-based feature extraction.

Internship corpus. Due to the lack of a large, structured, publicly available internship dataset, a small curated dictionary of internship postings (skills and requirements) was created manually to serve as the candidate corpus for matching.

Vectorization and matching. Both resumes and internship postings are converted into TF-IDF vectors, representing each document as a weighted bag of terms based on term frequency and

inverse document frequency. Internship postings are then ranked for a given resume by cosine similarity between the resume vector and each posting vector, with higher similarity indicating a stronger skill match.

Interface. The system is exposed through a Flask web application, allowing a resume to be submitted and ranked internship matches to be returned.

3. Evaluation

Formal evaluation (e.g., precision/recall against labeled ground-truth matches) was not conducted. The system was tested qualitatively by submitting multiple different resumes and inspecting the returned rankings; in these tests, the top-ranked postings consistently reflected the skills present in each resume, suggesting the matching behaved as intended on the scale tested. No quantitative metrics were computed, and the system has not been tested against a large-scale or external internship dataset, so its behavior at scale and against arbitrary, automatically scraped postings remains unverified.

4. Limitations

- **Dataset size and scope:** The internship corpus was manually constructed and small. Results have not been validated against a larger, real-world dataset of postings.
- **No quantitative evaluation:** Match quality was assessed by manual inspection rather than a defined metric or labeled benchmark.
- **TF-IDF's semantic limitations:** As a bag-of-words method, TF-IDF cannot capture synonymy or semantic similarity (e.g., "ML" vs. "Machine Learning") unless terms literally co-occur, which can cause relevant matches to be missed or irrelevant ones to be ranked highly.
- **Resume parsing robustness:** Parsing accuracy across varied resume formats and layouts was not stress-tested.

5. Future Work

Future iterations could incorporate semantic embeddings (e.g., sentence-transformers) in place of or alongside TF-IDF to better capture skill similarity beyond exact term overlap, evaluate the system against a larger and more realistic internship dataset, and introduce a quantitative evaluation framework (e.g., manually labeled relevance judgments) to measure match quality more rigorously.

6. Conclusion

This project demonstrates a working content-based internship recommendation pipeline using established NLP techniques (TF-IDF, cosine similarity) for resume-to-posting matching. While

preliminary and tested only on a small, manually curated dataset, the approach produced qualitatively sensible matches and serves as a foundation for future work involving larger datasets and more rigorous evaluation.